

| Data Validation          |     |     |       |
|--------------------------|-----|-----|-------|
| Features                 | Min | Max | Range |
| Age                      | 16  | 59  | 43    |
| Listening Time (minutes) | 10  | 299 | 289   |
| Songs Played per Day     | 1   | 99  | 98    |
| Skip Rate (%)            | 0   | 0.6 | 0.6   |
| Ads Listened per Week    | 0   | 49  | 49    |

| Missing Value Analysis & Duplicates Check |   |
|-------------------------------------------|---|
| Duplicates                                | 0 |
| Is Null                                   | 0 |

| Explanations & Notes                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <ul style="list-style-type: none"> <li>- Due to lag, I am starting the data cleaning process over using MySQL from Excel</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                      |
| <ul style="list-style-type: none"> <li>- Checking for null values so that my data is not misleading or miscalculated</li> <li>- Checking for null values in each column of the dataset using COUNT(CASE statements with no ELSE. I did not use the more general syntax COALESCE because it will change my null to 0 value. So when I count it will count the null and not null.</li> <li>- I've learned I can use SUM(CASE with ELSE (if i needed an else)</li> <li>- no nulls found in dataset (no bias kpi)</li> </ul> |
| <ul style="list-style-type: none"> <li>- Validating the ranges and values in numeric behavioral columns to make sure the data is valid and usable</li> <li>- Skip rate is a ratio (0.6) so this plays a part in formula for how we calculate the skip rate</li> <li>- MIN, MAX, AVG</li> </ul>                                                                                                                                                                                                                           |